

# ETL and ELT pipelines for Gaming Dataset

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# Table of Contents

|                                                                                                    |           |
|----------------------------------------------------------------------------------------------------|-----------|
| <b>Chapter 1: Introduction .....</b>                                                               | <b>4</b>  |
| 1.1 Background and motivation .....                                                                | 4         |
| 1.2 Problem statement .....                                                                        | 5         |
| 1.3 Objectives of the Data engineering Project .....                                               | 5         |
| 1.4 Sustainable Development Goals (SDGs) .....                                                     | 6         |
| <b>Chapter 2 : System Design and Architecture .....</b>                                            | <b>6</b>  |
| 2.1 Data sources and collection methods .....                                                      | 6         |
| ETL Pipeline .....                                                                                 | 6         |
| ELT Pipeline .....                                                                                 | 7         |
| 2.2 Data Ingestion method .....                                                                    | 7         |
| ETL Pipeline .....                                                                                 | 7         |
| ELT Pipeline .....                                                                                 | 8         |
| 2.3 Data cleaning and transformation methods .....                                                 | 8         |
| ETL Pipeline .....                                                                                 | 8         |
| ELT Pipeline .....                                                                                 | 9         |
| <b>Chapter 3: ETL Pipeline Design .....</b>                                                        | <b>9</b>  |
| 3.1 Data pipeline architectute(ETL) .....                                                          | 9         |
| 3.2 Extract: Collecting data from sources (APIs, databases, files) .....                           | 10        |
| 3.3 Transform: Cleaning, filtering, aggregating data before loading .....                          | 11        |
| 3.4 Load: Storing transformed data into a data warehouse (e.g.,Sqlite ) .....                      | 13        |
| <b>Chapter 4: Implementation and Testing .....</b>                                                 | <b>15</b> |
| 4.1 Step-by-step ETL process execution using python program .....                                  | 15        |
| 4.2 Visualize using BI dashboard .....                                                             | 18        |
| .....                                                                                              | 18        |
| 4.3 Performance challenges and solutions .....                                                     | 18        |
| <b>Chapter 5: (Phase 2) Introduction to ELT .....</b>                                              | <b>19</b> |
| 5.1 Difference between ELT (Extract, Load, Transform) and ETL (Extract ,Transform ,<br>load) ..... | 19        |
| 5.2 Why ELT is used .....                                                                          | 19        |
| 5.3 Tools used as Datawarehouse and libraries for ELT .....                                        | 19        |
| <b>Chapter 6: ELT Pipeline Design .....</b>                                                        | <b>21</b> |
| 6.1 Data pipeline architecture (ELT) .....                                                         | 21        |
| 6.2 Data sources: Collecting raw data .....                                                        | 21        |
| 6.3 Load the raw data in datawarehouse .....                                                       | 23        |

|                                                                     |           |
|---------------------------------------------------------------------|-----------|
| 6.4 Transform: SQL based transformation .....                       | 25        |
| <b>Chapter 7: Implementation &amp; Results .....</b>                | <b>29</b> |
| 7.1 Execution of ELT workflow .....                                 | 29        |
| 7.2 Visualize using BI dashboard .....                              | 31        |
| 7.3 Automation of ELT pipeline using Task scheduler .....           | 31        |
| 7.4 Comparison of ETL vs. ELT (speed, scalability, efficiency)..... | 37        |
| 7.5 Conclusion .....                                                | 39        |
| 7.6 References (IEEE format) .....                                  | 40        |

## **Chapter 1: Introduction**

Data engineering is important in modern systems because companies collect large amounts of data from different sources. Raw data is usually not ready for analysis because it may contain missing values, duplicated records, or inconsistent formats. Therefore, data pipelines are used to extract, clean, transform, and store data in an organized way.[1]

In this project, we implemented both ETL (Extract, Transform, Load) and ELT (Extract, Load, Transform) pipelines using a video games dataset. The dataset includes information such as game names, platforms, genres, publishers, release years, and global sales. We also used the RAWG API to collect additional information like ratings, Metacritic scores, and playtime.

The main purpose of this project is to compare ETL and ELT pipelines in terms of performance, scalability, efficiency, and data quality using practical implementation with Python, SQLite, PostgreSQL, and SQL.

### **1.1 Background and motivation**

Data engineering helps organizations prepare raw data for analysis and decision-making. Modern systems collect data from APIs, databases, and CSV files, but this data is often incomplete or inconsistent. Because of this, data cleaning and transformation are necessary before using data analysis.

The gaming industry produces a large amount of data related to game sales, ratings, publishers, and platforms. This data can provide useful insights, but it must first be processed and organized correctly.

In this project, we implemented ETL and ELT pipelines to understand how each approach handles data extraction, transformation, validation, and storage. We also compared the advantages and limitations of both methods to identify the best use cases for each one.

## **1.2 Problem statement**

As data volume increases, organizations face challenges in processing and integrating data from multiple sources. It is also difficult to decide whether ETL or ELT is more suitable for handling large datasets and different data formats.[2]

The video games dataset used in this project contains missing values, duplicate records, inconsistent formats, and incomplete information collected from CSV files and APIs. These problems reduce data quality and affect the accuracy of analysis.

Therefore, this project focuses on designing ETL and ELT pipelines that can extract, clean, transform, validate, and store gaming data in a structured format suitable for analysis and visualization.

**1.3 Objectives of the Data engineering Project** The objectives of this project are:

- Design and implement an ETL pipeline for video games data.
- Design and implement an ELT pipeline using PostgreSQL.
- Extract gaming data from CSV files, APIs, and databases.
- Clean and transform the data to improve quality and consistency.
- Store processed data in SQLite and PostgreSQL databases.
- Compare ETL and ELT in terms of performance, scalability, and efficiency.
- Visualize the processed data using BI dashboards.

## **1.4 Sustainable Development Goals (SDGs)**

This project supports SDG 9: Industry, Innovation, and Infrastructure [3] by implementing modern data engineering technologies such as ETL and ELT pipelines, APIs, databases, and cloud data warehouses to improve data processing and management.

The project also supports SDG 8: Decent Work and Economic Growth [3] by improving data workflows and enabling efficient data analysis for better decision-making.

## **Chapter 2 : System Design and Architecture**

This chapter explains the system architecture used to implement the ETL and ELT pipelines. The project combines multiple data sources and applies ingestion, cleaning, transformation, and storage processes to prepare the data for analysis.

### **2.1 Data sources and collection methods**

In this project, different data sources were used for ETL and ELT pipelines.

#### **ETL Pipeline**

Two main sources were used:

- CSV file (Video\_Games\_500.csv):  
This dataset contains information about video games such as game names, platforms, genres, publishers, release years, and global sales.
- RAWG API:  
The API was used to collect additional game details such as ratings, Metacritic scores, release dates, and playtime.

## **ELT Pipeline**

The following sources were used:

- SQLite Database:  
Contains developer information such as developer names, headquarters, and foundation years.
- CSV file (Video\_Games.csv):  
Contains video games data including platform, genre, publisher, and global sales.

These sources were selected to simulate real-world data engineering scenarios using APIs, CSV files, and relational databases.

## **2.2 Data Ingestion method**

In this project, batch ingestion was used to collect and process data.

### **ETL Pipeline**

- Data was extracted from the CSV file using pandas.
- Additional data was collected from the RAWG API using requests.
- The datasets were merged and transformed before loading into SQLite.

### **ELT Pipeline**

- Raw data from the SQLite database and CSV file were loaded directly into PostgreSQL staging tables.
- Data transformation was performed later inside PostgreSQL using SQL queries.

## **2.3 Data cleaning and transformation methods**

Several cleaning and transformation methods were applied in both ETL and ELT pipelines.

### **ETL Pipeline**

The data was cleaned before loading using Python and pandas. The following steps were applied:

- Cleaning text values and removing extra spaces.
- Replacing invalid text values with missing values.
- Handling missing values by removing rows with missing critical columns and filling non-critical values.
- Converting numeric columns into suitable numeric data types.
- Converting release year values into integer format.
- Removing duplicate records based on game name and platform.
- Merging CSV and RAWG API data into one dataset using a left join.
- Resetting the dataset index after cleaning.

After transformation, the cleaned dataset was loaded into SQLite.

### **ELT Pipeline**

In the ELT workflow, raw data was loaded first into PostgreSQL, and transformation was done later using SQL queries.

The transformations included:

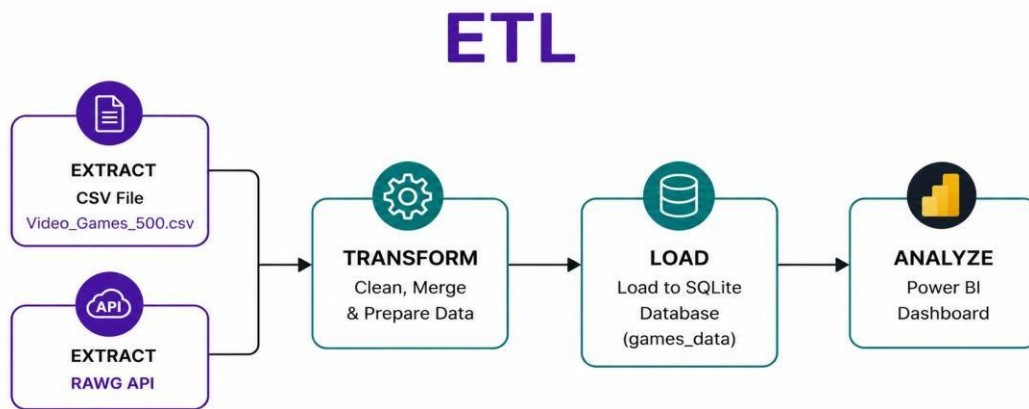
- Removing duplicate records using `SELECT DISTINCT`.
- Cleaning text using `TRIM()`.
- Standardizing platform names using `UPPER()`.

- Replacing missing values using COALESCE().
- Converting release year values into integers using CAST().

These transformations improved data quality and prepared the final tables for dashboard analysis and visualization.

## Chapter 3: ETL Pipeline Design

### 3.1 Data pipeline architecture(ETL)



*Figure 1 ETL pipeline architecture*

### 3.2 Extract: Collecting data from sources (APIs, databases, files)

In this phase, data was collected from two sources:

- A CSV file (**Video\_Games\_500\_dirty.csv**) containing video games data.
- The **RAWG API** for additional game information such as ratings and release dates.

Python libraries such as pandas, requests, and sqlite3 were used during the extraction process

```

ETL pipeline.py > extract_api
1  import pandas as pd
2  import requests
3  import sqlite3
4
5  # Settings
6  CSV_FILE_PATH = "Video_Games_500_dirty.csv"
7  DB_PATH = "new_gaming_project.db"
8  API_KEY = "053cf90e3e444d15acad6f654dc04013"
9

```

*Figure 2 Import data*

```

11 # --- 1. Extract CSV data ---
12 def extract_csv(file_path):
13     df = pd.read_csv(file_path)
14
15     # Clean column names
16     df.columns = df.columns.str.strip().str.upper()
17
18     print("CSV extracted.")
19     return df
20

```

*Figure 3 Extract CSV*

```

22 # --- 2. Extract API data ---
23 def extract_api(names):
24     results = []
25
26     for name in names:
27         url = "https://api.rawg.io/api/games"
28
29         params = {
30             "key": API_KEY,
31             "search": name,
32             "page_size": 1
33         }
34
35         try:
36             response = requests.get(url, params=params, timeout=10)
37             data = response.json()
38
39             if data.get("results"):
40                 game = data["results"][0]
41
42                 results.append({
43                     "TITLE": name,
44                     "RAWG_NAME": game.get("name"),
45                     "RAWG_RATING": game.get("rating"),
46                     "METACRITIC": game.get("metacritic"),
47                     "RAWG_RELEASED": game.get("released"),
48                     "RAWG_PLAYTIME": game.get("playtime")
49                 })
50
51         except:
52             print("API skipped:", name)
53
54     print("API extracted.")
55     return pd.DataFrame(results)
56

```

*Figure 4 Extract API*

### 3.3 Transform: Cleaning, filtering, aggregating data before loading

After extraction, several transformation steps were applied to clean and prepare the data before loading:

- Missing values in ratings, scores, and playtime columns were replaced with suitable values.
- Numeric columns were converted into appropriate data types.
- Duplicate rows were removed based on game name and platform.
- The release year column was converted into integer format.
- Data from the CSV file and RAWG API were merged into one unified dataset.
- The dataset index was reset after cleaning to maintain proper structure.

Finally, the cleaned and merged dataset was prepared successfully before loading into the SQLite database.

```
ETL pipeline.py > transform
59 def transform(csv_df, api_df):
147     # Fill API numeric values only with 0 because API may not return them
148     api_numeric_fill = [
149         "RAWG_RATING",
150         "METACRITIC",
151         "RAWG_PLAYTIME"
152     ]
153
154     for col in api_numeric_fill:
155         if col in df.columns:
156             df[col] = df[col].fillna(0)
157
158     # Fill optional score/count values with 0
159     optional_numeric_fill = [
160         "CRITIC_SCORE",
161         "CRITIC_COUNT",
162         "USER_SCORE",
163         "USER_COUNT"
164     ]
165
166     for col in optional_numeric_fill:
167         if col in df.columns:
168             df[col] = df[col].fillna(0)
169
170     # Convert year to integer
171     df["YEAR_OF_RELEASE"] = df["YEAR_OF_RELEASE"].astype(int)
172
173     # Remove duplicates
174     df.drop_duplicates(
175         subset=["NAME", "PLATFORM"],
176         inplace=True
177     )
178
179     # Reset index
180     df.reset_index(drop=True, inplace=True)
```

Figure 5 Data Transformation

| A                                                         | B        | C               | D            | E                          | F        | G        | H        | I           | J            | K            | L            | M          | N          | O                     | P      | Q        | R        | S        | T           | U            | V            | W            | X          | Y          | Z                     |        |
|-----------------------------------------------------------|----------|-----------------|--------------|----------------------------|----------|----------|----------|-------------|--------------|--------------|--------------|------------|------------|-----------------------|--------|----------|----------|----------|-------------|--------------|--------------|--------------|------------|------------|-----------------------|--------|
| Name                                                      | Platform | Year of Release | Genre        | Publisher                  | NA Sales | EU Sales | JP Sales | Other Sales | Global Sales | Critic Score | Critic Count | User Score | User Count | Developer             | Rating | NA Sales | EU Sales | JP Sales | Other Sales | Global Sales | Critic Score | Critic Count | User Score | User Count | Developer             | Rating |
| The Incredible Hulk Ultimate Destruction                  | PS2      | 2005            | Action       | Vivendi Games              | 0.17     | 0.05     | 0.01     | 0.01        | 0.23         | 54           | 47           | 8          | 22         | Radical Entertainment | E      | 0.17     | 0.05     | 0.01     | 0.01        | 0.23         | 54           | 47           | 8          | 22         | Radical Entertainment | E      |
| Datenen no Ama Yawaku Kaikan Phrase                       | DS       | 2010            | Adventure    | Fufu                       | 0        | 0        | 0.01     | 0           | 0.01         | 0            | 0            | 0          | 0          | 2K Games              | E      | 0        | 0        | 0.01     | 0           | 0.01         | 0            | 0            | 0          | 0          | 2K Games              | E      |
| NSA-2K17                                                  | X360     | 2016            | Sports       | Take-Two Interactive       | 0.09     | 0.02     | 0        | 0.01        | 0.12         | 0            | 0            | 0          | 0          | EA Canada             | T      | 0.09     | 0.02     | 0        | 0.01        | 0.12         | 0            | 0            | 0          | 0          | EA Canada             | T      |
| Scoddy Do! Unmasked                                       | DS       | 2005            | Platform     | THQ                        | 0.07     | 0        | 0        | 0.01        | 0.09         | 0            | 0            | 0          | 0          | Ubisoft               | E      | 0.07     | 0        | 0        | 0.01        | 0.09         | 0            | 0            | 0          | 0          | Ubisoft               | E      |
| Left Brain Right Brain: Use Both Hands Than Both Sides    | DS       | 2007            | Misc         | 2005 Games                 | 0.15     | 0        | 0        | 0.01        | 0.16         | 0            | 0            | 0          | 0          | DC Studios            | E      | 0.15     | 0        | 0        | 0.01        | 0.16         | 0            | 0            | 0          | 0          | DC Studios            | E      |
| Zehn Bell Electric Arena                                  | GBA      | 2003            | Fighting     | Banpresto                  | 0.02     | 0.01     | 0        | 0           | 0.03         | 0            | 0            | 0          | 0          | EA Canada             | T      | 0.02     | 0.01     | 0        | 0           | 0.03         | 0            | 0            | 0          | 0          | EA Canada             | T      |
| Disney's Chicken Little: Ace in Action                    | ds       | 2006            | Shooter      | Disney Interactive Studios | 0.06     | 0        | 0        | 0           | 0.06         | 66           | 13           | 8d         | 5          | Two Tribes            | E      | 0.06     | 0        | 0        | 0           | 0.06         | 66           | 13           | 8d         | 5          | Two Tribes            | E      |
| Cabotown                                                  | XB       | 2004            | Action       | Electronic Arts            | 0.04     | 0.01     | 0        | 0           | 0.06         | 45           | 33           | 4.6        | 9          | Team 17               | E      | 0.04     | 0.01     | 0        | 0           | 0.06         | 45           | 33           | 4.6        | 9          | Team 17               | E      |
| Rubik's World                                             | Wii      | 2008            | Puzzle       | Game Factory               | 0.12     | 0.01     | 0        | 0.01        | 0.14         | 64           | 9            | 8d         | 0          | Ubisoft               | E      | 0.12     | 0.01     | 0        | 0.01        | 0.14         | 64           | 9            | 8d         | 0          | Ubisoft               | E      |
| Super Robot Taisen OG Original Generations Gaiden         | PS2      | 2007            | Strategy     | Banpresto                  | 0        | 0        | 0.3      | 0           | 0.3          | 0            | 0            | 0          | 0          | The Sims Studio       | T      | 0        | 0        | 0.3      | 0           | 0.3          | 0            | 0            | 0          | 0          | The Sims Studio       | T      |
| J League Pro Soccer Club o Takarun! 7 Euro Plus           | PS2      | 2011            | Sports       | Sega                       | 0        | 0        | 0.21     | 0           | 0.21         | 0            | 0            | 0          | 0          | Bethesda Softworks    | M      | 0        | 0        | 0.21     | 0           | 0.21         | 0            | 0            | 0          | 0          | Bethesda Softworks    | M      |
| Maj de Marabou LEC de Ukuru - DS Hsibu Boku 3-Kyuu        | DS       | 2008            | Misc         | Square Enix                | 0        | 0        | 0.02     | 0           | 0.02         | 0            | 0            | 0          | 0          | Ubisoft               | E      | 0        | 0        | 0.02     | 0           | 0.02         | 0            | 0            | 0          | 0          | Ubisoft               | E      |
| The Sims 3: Town Life Stuff                               | PC       | 2011            | Simulation   | Electronic Arts            | 0.11     | 0.17     | 0        | 0.05        | 0.34         | 94           | 54           | 8.1        | 2454       | EA Canada             | T      | 0.11     | 0.17     | 0        | 0.05        | 0.34         | 94           | 54           | 8.1        | 2454       | EA Canada             | T      |
| The Elder Scrolls IV: Oblivion                            | PC       | 2006            | Role-Playing | Take-Two Interactive       | 0.01     | 0.2      | 0        | 0.04        | 0.25         | 94           | 54           | 8.1        | 2454       | Ubisoft               | E      | 0.01     | 0.2      | 0        | 0.04        | 0.25         | 94           | 54           | 8.1        | 2454       | Ubisoft               | E      |
| Let's Chess                                               | X360     | 2011            | Misc         | Take-Two Interactive       | 0.12     | 0        | 0        | 0.01        | 0.14         | 0            | 0            | 0          | 0          | Next Level Games      | T      | 0.12     | 0        | 0        | 0.01        | 0.14         | 0            | 0            | 0          | 0          | Next Level Games      | T      |
| Ramen 2: Revolution                                       | PS2      | 2000            | Platform     | Ubisoft                    | 0.15     | 0.11     | 0        | 0.04        | 0.3          | 61           | 34           | 8.2        | 42         | EA Canada             | T      | 0.15     | 0.11     | 0        | 0.04        | 0.3          | 61           | 34           | 8.2        | 42         | EA Canada             | T      |
| Captain America: Super Soldier                            | PS3      | 2011            | Action       | Sega                       | 0.1      | 0.06     | 0        | 0.03        | 0.19         | 61           | 34           | 8.2        | 42         | Traveler's Tales      | E10+   | 0.1      | 0.06     | 0        | 0.03        | 0.19         | 61           | 34           | 8.2        | 42         | Traveler's Tales      | E10+   |
| NSA Live 08                                               | PS2      | 2007            | Sports       | Electronic Arts            | 0.59     | 0.02     | 0.01     | 0.1         | 0.72         | 68           | 10           | 6.9        | 9          | Team 17               | E      | 0.59     | 0.02     | 0.01     | 0.1         | 0.72         | 68           | 10           | 6.9        | 9          | Team 17               | E      |
| LEGO Star Wars II: The Original Trilogy                   | XB       | 2006            | Action       | LucasArts                  | 0.33     | 0.1      | 0        | 0.02        | 0.44         | 85           | 27           | 8.6        | 15         | Ubisoft               | E      | 0.33     | 0.1      | 0        | 0.02        | 0.44         | 85           | 27           | 8.6        | 15         | Ubisoft               | E      |
| Minicraft: Story Mode                                     | X360     | 2015            | Adventure    | Mojang                     | 0.48     | 0.33     | 0        | 0.08        | 0.88         | 0            | 0            | 0          | 0          | Team 17               | E      | 0.48     | 0.33     | 0        | 0.08        | 0.88         | 0            | 0            | 0          | 0          | Team 17               | E      |
| Hatsunaku! Reineroku                                      | PS2      | 2010            | Adventure    | Idea Factory               | 0        | 0        | 0.05     | 0           | 0.05         | 0            | 0            | 0          | 0          | Team 17               | E      | 0        | 0        | 0.05     | 0           | 0.05         | 0            | 0            | 0          | 0          | Team 17               | E      |
| Worms Forts Under Siege                                   | XB       | 2004            | Strategy     | Sega                       | 0        | 0.01     | 0        | 0           | 0.07         | 67           | 15           | 8          | 6          | Team 17               | E      | 0        | 0.01     | 0        | 0           | 0.07         | 67           | 15           | 8          | 6          | Team 17               | E      |
| Gundam Battle Online                                      | DC       | 2001            | Action       | Namco Bandai Games         | 0        | 0        | 0.04     | 0           | 0.04         | 0            | 0            | 0          | 0          | Team 17               | E      | 0        | 0        | 0.04     | 0           | 0.04         | 0            | 0            | 0          | 0          | Team 17               | E      |
| Pokemon Omega Ruby/Pokemon Alpha Sapphire                 | 3DS      | 2014            | Role-Playing | Nintendo                   | 4.35     | 3.49     | 3.1      | 0.74        | 11.68        | 0            | 0            | 0          | 0          | Team 17               | E      | 4.35     | 3.49     | 3.1      | 0.74        | 11.68        | 0            | 0            | 0          | 0          | Team 17               | E      |
| London 2012 The Official Video Games of the Olympic Games | X360     | 2012            | Sports       | Sega                       | 0.07     | 0        | 0        | 0.07        | 0.56         | 0            | 0            | 0          | 0          | Team 17               | E      | 0.07     | 0        | 0        | 0.07        | 0.56         | 0            | 0            | 0          | 0          | Team 17               | E      |
| Kid's Adventure                                           | NES      | 1983            | Platform     | Ubisoft                    | 0.78     | 0.14     | 0.8      | 0.02        | 1.75         | 0            | 0            | 0          | 0          | Team 17               | E      | 0.78     | 0.14     | 0.8      | 0.02        | 1.75         | 0            | 0            | 0          | 0          | Team 17               | E      |
| Let's Chess                                               | PS3      | 2011            | Action       | Sega                       | 0.1      | 0.06     | 0        | 0.03        | 0.19         | 61           | 34           | 8.2        | 42         | Team 17               | E      | 0.1      | 0.06     | 0        | 0.03        | 0.19         | 61           | 34           | 8.2        | 42         | Team 17               | E      |
| NSA Live 08                                               | PS2      | 2007            | Sports       | Electronic Arts            | 0.59     | 0.02     | 0.01     | 0.1         | 0.72         | 68           | 10           | 6.9        | 9          | Team 17               | E      | 0.59     | 0.02     | 0.01     | 0.1         | 0.72         | 68           | 10           | 6.9        | 9          | Team 17               | E      |
| LEGO Star Wars II: The Original Trilogy                   | XB       | 2006            | Action       | LucasArts                  | 0.33     | 0.1      | 0        | 0.02        | 0.44         | 85           | 27           | 8.6        | 15         | Team 17               | E      | 0.33     | 0.1      | 0        | 0.02        | 0.44         | 85           | 27           | 8.6        | 15         | Team 17               | E      |
| Minicraft: Story Mode                                     | X360     | 2015            | Adventure    | Mojang                     | 0.48     | 0.33     | 0        | 0.08        | 0.88         | 0            | 0            | 0          | 0          | Team 17               | E      | 0.48     | 0.33     | 0        | 0.08        | 0.88         | 0            | 0            | 0          | 0          | Team 17               | E      |
| Hatsunaku! Reineroku                                      | PS2      | 2010            | Adventure    | Idea Factory               | 0        | 0        | 0.05     | 0           | 0.05         | 0            | 0            | 0          | 0          | Team 17               | E      | 0        | 0        | 0.05     | 0           | 0.05         | 0            | 0            | 0          | 0          | Team 17               | E      |
| Worms Forts Under Siege                                   | XB       | 2004            | Strategy     | Sega                       | 0        | 0.01     | 0        | 0           | 0.07         | 67           | 15           | 8          | 6          | Team 17               | E      | 0        | 0.01     | 0        | 0           | 0.07         | 67           | 15           | 8          | 6          | Team 17               | E      |
| Gundam Battle Online                                      | DC       | 2001            | Action       | Namco Bandai Games         | 0        | 0        | 0.04     | 0           | 0.04         | 0            | 0            | 0          | 0          | Team 17               | E      | 0        | 0        | 0.04     | 0           | 0.04         | 0            | 0            | 0          | 0          | Team 17               | E      |
| Pokemon Omega Ruby/Pokemon Alpha Sapphire                 | 3DS      | 2014            | Role-Playing | Nintendo                   | 4.35     | 3.49     | 3.1      | 0.74        | 11.68        | 0            | 0            | 0          | 0          | Team 17               | E      | 4.35     | 3.49     | 3.1      | 0.74        | 11.68        | 0            | 0            | 0          | 0          | Team 17               | E      |
| London 2012 The Official Video Games of the Olympic Games | X360     | 2012            | Sports       | Sega                       | 0.07     | 0        | 0        | 0.07        | 0.56         | 0            | 0            | 0          | 0          | Team 17               | E      | 0.07     | 0        | 0        | 0.07        | 0.56         | 0            | 0            | 0          | 0          | Team 17               | E      |
| Kid's Adventure                                           | NES      | 1983            | Platform     | Ubisoft                    | 0.78     | 0.14     | 0.8      | 0.02        | 1.75         | 0            | 0            | 0          | 0          | Team 17               | E      | 0.78     | 0.14     | 0.8      | 0.02        | 1.75         | 0            | 0            | 0          | 0          | Team 17               | E      |
| Let's Chess                                               | PS3      | 2011            | Action       | Sega                       | 0.1      | 0.06     | 0        | 0.03        | 0.19         | 61           | 34           | 8.2        | 42         | Team 17               | E      | 0.1      | 0.06     | 0        | 0.03        | 0.19         | 61           | 34           | 8.2        | 42         | Team 17               | E      |
| NSA Live 08                                               | PS2      | 2007            | Sports       | Electronic Arts            | 0.59     | 0.02     | 0.01     | 0.1         | 0.72         | 68           | 10           | 6.9        | 9          | Team 17               | E      | 0.59     | 0.02     | 0.01     | 0.1         | 0.72         | 68           | 10           | 6.9        | 9          | Team 17               | E      |
| LEGO Star Wars II: The Original Trilogy                   | XB       | 2006            | Action       | LucasArts                  | 0.33     | 0.1      | 0        | 0.02        | 0.44         | 85           | 27           | 8.6        | 15         | Team 17               | E      | 0.33     | 0.1      | 0        | 0.02        | 0.44         | 85           | 27           | 8.6        | 15         | Team 17               | E      |
| Minicraft: Story Mode                                     | X360     | 2015            | Adventure    | Mojang                     | 0.48     | 0.33     | 0        | 0.08        | 0.88         | 0            | 0            | 0          | 0          | Team 17               | E      | 0.48     | 0.33     | 0        | 0.08        | 0.88         | 0            | 0            | 0          | 0          | Team 17               | E      |
| Hatsunaku! Reineroku                                      | PS2      | 2010            | Adventure    | Idea Factory               | 0        | 0        | 0.05     | 0           | 0.05         | 0            | 0            | 0          | 0          | Team 17               | E      | 0        | 0        | 0.05     | 0           | 0.05         | 0            | 0            | 0          | 0          | Team 17               | E      |
| Worms Forts Under Siege                                   | XB       | 2004            | Strategy     | Sega                       | 0        | 0.01     | 0        | 0           | 0.07         | 67           | 15           | 8          | 6          | Team 17               | E      | 0        | 0.01     | 0        | 0           | 0.07         | 67           | 15           | 8          | 6          | Team 17               | E      |
| Gundam Battle Online                                      | DC       | 2001            | Action       | Namco Bandai Games         | 0        | 0        | 0.04     | 0           | 0.04         | 0            | 0            | 0          | 0          | Team 17               | E      | 0        | 0        | 0.04     | 0           | 0.04         | 0            | 0            | 0          | 0          | Team 17               | E      |
| Pokemon Omega Ruby/Pokemon Alpha Sapphire                 | 3DS      | 2014            | Role-Playing | Nintendo                   | 4.35     | 3.49     | 3.1      | 0.74        | 11.68        | 0            | 0            | 0          | 0          | Team 17               | E      | 4.35     | 3.49     | 3.1      | 0.74        | 11.68        | 0            | 0            | 0          | 0          | Team 17               | E      |
| London 2012 The Official Video Games of the Olympic Games | X360     | 2012            | Sports       | Sega                       | 0.07     | 0        | 0        | 0.07        | 0.56         | 0            | 0            | 0          | 0          | Team 17               | E      | 0.07     | 0        | 0        | 0.07        | 0.56         | 0            | 0            | 0          | 0          | Team 17               | E      |
| Kid's Adventure                                           | NES      | 1983            | Platform     | Ubisoft                    | 0.78     | 0.14     | 0.8      | 0.02        | 1.75         | 0            | 0            | 0          | 0          | Team 17               | E      | 0.78     | 0.14     | 0.8      | 0.02        | 1.75         | 0            | 0            | 0          | 0          | Team 17               | E      |
| Let's Chess                                               | PS3      | 2011            | Action       | Sega                       | 0.1      | 0.06     | 0        | 0.03        | 0.19         | 61           | 34           | 8.2        | 42         | Team 17               | E      | 0.1      | 0.06     | 0        | 0.03        | 0.19         | 61           | 34           | 8.2        | 42         | Team 17               | E      |
| NSA Live 08                                               | PS2      | 2007            | Sports       | Electronic Arts            | 0.59     | 0.02     | 0.01     | 0.1         | 0.72         | 68           | 10           | 6.9        | 9          | Team 17               | E      | 0.59     | 0.02     | 0.01     | 0.1         | 0.72         | 68           | 10           | 6.9        | 9          | Team 17               | E      |
| LEGO Star Wars II: The Original Trilogy                   | XB       | 2006            | Action       | LucasArts                  | 0.33     | 0.1      | 0        | 0.02        | 0.44         | 85           | 27           | 8.6        | 15         | Team 17               | E      | 0.33     | 0.1      | 0        | 0.02        | 0.44         | 85           | 27           | 8.6        | 15         | Team 17               | E      |
| Minicraft: Story Mode                                     | X360     | 2015            | Adventure    | Mojang                     | 0.48     | 0.33     | 0        | 0.08        | 0.88         | 0            | 0            | 0          | 0          | Team 17               | E      | 0.48     | 0.33     | 0        | 0.08        | 0.88         | 0            | 0            | 0          | 0          | Team 17               | E      |
| Hatsunaku! Reineroku                                      | PS2      | 2010            | Adventure    | Idea Factory               | 0        | 0        | 0.05     | 0           | 0.05         | 0            | 0            | 0          | 0          | Team 17               | E      | 0        | 0        | 0.05     | 0           | 0.05         | 0            | 0            | 0          | 0          | Team 17               | E      |
| Worms Forts Under Siege                                   | XB       | 2004            | Strategy     | Sega                       | 0        | 0.01     | 0        | 0           | 0.07         | 67           | 15           | 8          | 6          | Team 17               | E      | 0        | 0.01     | 0        | 0           | 0.07         | 67           | 15           | 8          | 6          | Team 17               | E      |
| Gundam Battle Online                                      | DC       | 2001            | Action       | Namco Bandai Games         | 0        | 0        | 0.04     | 0           | 0.04         | 0            | 0            | 0          | 0          | Team 17               | E      | 0        | 0        | 0.04     | 0           | 0.04         | 0            | 0            | 0          | 0          | Team 17               | E      |
| Pokemon Omega Ruby/Pokemon Alpha Sapphire                 | 3DS      | 2014            | Role-Playing | Nintendo                   | 4.35     | 3.49     | 3.1      | 0.74        | 11.68        | 0            | 0            | 0          | 0          | Team 17               | E      | 4.35     | 3.49     | 3.1      | 0.74        | 11.68        | 0            | 0            | 0          | 0          | Team 17               | E      |
| London 2012 The Official Video Games of the Olympic Games | X360     | 2012            | Sports       | Sega                       | 0.07     | 0        | 0        | 0.07        | 0.56         | 0            | 0            | 0          | 0          | Team 17               | E      | 0.07     | 0        | 0        | 0.07        | 0.56         | 0            | 0            | 0          | 0          | Team 17               | E      |
| Kid's Adventure                                           | NES      | 1983            | Platform     | Ubisoft                    | 0.78     | 0.14     | 0.8      | 0.02        | 1.75         | 0            | 0            | 0          | 0          | Team 17               | E      | 0.78     | 0.14     | 0.8      | 0.02        | 1.75         | 0            | 0            | 0          | 0          | Team 17               | E      |
| Let's Chess                                               | PS3      | 2011            | Action       | Sega                       | 0.1      | 0.06     | 0        | 0.03        | 0.19         | 61           | 34           | 8.2        | 42         | Team 17               | E      | 0.1      | 0.06     | 0        | 0.03        | 0.19         | 61           | 34           | 8.2        | 42         | Team 17               | E      |
| NSA Live 08                                               | PS2      | 2007            | Sports       | Electronic Arts            | 0.59     | 0.02     | 0.01     | 0.1         | 0.72         | 68           | 10           | 6.9        | 9          | Team 17               | E      | 0.59     | 0.02     | 0.01     | 0.1         | 0.72         | 68           | 10           | 6.9        | 9          | Team 17               | E      |
| LEGO Star Wars II: The Original Trilogy                   | XB       | 2006            |              |                            |          |          |          |             |              |              |              |            |            |                       |        |          |          |          |             |              |              |              |            |            |                       |        |

### 3.4 Load: Storing transformed data into a data warehouse (e.g.,Sqlite )

After completing the transformation process, the final cleaned dataset was loaded into an SQLite database named **new\_gaming\_project.db**. The transformed data was stored in a table called **games\_data**, which contains the unified dataset collected from both the CSV file and the RAWG API.

The data was loaded using the `to_sql()` function with the `replace` option to ensure the table is updated whenever the ETL pipeline is executed again.

Storing the data in SQLite provides a structured format that supports efficient querying, data management, and visualization using tools such as Power BI.

```
195 # --- 4. Load data to SQLite database ---
196 def load(df, db):
197     conn = sqlite3.connect(db)
198
199     df.to_sql(
200         "games_data",
201         conn,
202         if_exists="replace",
203         index=False
204     )
205
206     conn.close()
207
208     print("Data loaded to Database.")
209
```

*Figure 8 Data Loading*

## Chapter 4: Implementation and Testing

### 4.1 Step-by-step ETL process execution using python program

Python was used to build the ETL (Extract, Transform, Load) pipeline to automate the process of extracting, cleaning, merging, and storing video games data from both the CSV dataset and the RAWG API.

### **Step 1: Extract Data from CSV:**

First, the dataset “Video\_Games\_500.csv” was loaded using pandas. The dataset includes information about video games such as game title, platform, genre, publisher, sales, and review scores. After importing the file, the column names were standardized to maintain consistency before merging.

### **Step 2: Extract Data from API:**

Then, additional game details were collected from the RAWG API using the game titles available in the CSV dataset. The API returned extra information including RAWG rating, Metacritic score, release date, and game playtime. The extracted API records were stored in a separate pandas DataFrame.

### **Step 3: Merge and Clean the Data:**

After extraction, the CSV data and API data were combined into a single unified DataFrame based on the game title. Several data cleaning techniques were applied, such as handling missing values, converting columns into appropriate numeric formats, removing duplicate records, and resetting the index to improve data quality and consistency.

### **Step 4: Load Cleaned Data to SQLite:**

Finally, the transformed dataset was loaded into an SQLite database named “new\_gaming\_project.db”. The final data was stored in the “games\_data” table, which can be used later for SQL queries, dashboard

```
C:\Users\danhs\venv\sqlite>sqlite3 "C:\Users\danhs\venv\new_gaming_project.db"
SQLite version 3.51.2 2026-01-09 17:27:48
Enter ".help" for usage hints.
sqlite> .tables
games_data
sqlite> .mode column
sqlite> .headers on
sqlite> SELECT * FROM games_data LIMIT 10;
```

| NAME                                                    | LES  | OTHER_SALES | GLOBAL_SALES                 | CRITIC_SCORE | CRITIC_COUNT | PLATFORM | YEAR_OF_RELEASE | GENRE                 | DEVELOPER                  | PUBLISHER     | RATING | TITLE                          | NA_SALES | EU_SALES | JP_SA |
|---------------------------------------------------------|------|-------------|------------------------------|--------------|--------------|----------|-----------------|-----------------------|----------------------------|---------------|--------|--------------------------------|----------|----------|-------|
|                                                         |      |             | RAWG_NAME                    |              |              |          | RAWG_RATING     | METACRITIC            | RAWG_RELEASED              | RAWG_PLAYTIME |        |                                |          |          |       |
| The Incredible Hulk: Ultimate Destruction               | 0.01 | 0.23        | 84.0                         | 47.0         | 8.0          | 2005     | 22.0            | Action                | Vivendi Games              |               | 0.17   | 0.05                           | 0.0      |          |       |
| Destruction                                             |      |             |                              |              |              |          |                 | Radical Entertainment | T                          |               |        | The Incredible Hulk: Ultimate  |          |          |       |
| Datenshi no Amai Yuuwaku x Kaikan Phrase                | 0.0  | 0.01        | 0.0                          | 0.0          | 0.0          | 2010     | 0.0             | Adventure             | FuRyu                      |               | 0.0    | 0.0                            | 0.01     |          |       |
| kan Phrase                                              |      |             | No. 10                       |              |              |          |                 | Unknown Developer     |                            | Not Rated     |        | Datenshi no Amai Yuuwaku x Kai |          |          |       |
| NBA 2K17                                                | 0.01 | 0.12        | 0.0                          | 0.0          | 3.4          | 2016     | 11.0            | sports                | Take-Two Interactive       |               | 0.09   | 0.02                           | 0.0      |          |       |
|                                                         |      |             | NBA 2K17                     |              |              |          |                 | 2K Games              | E                          |               |        | NBA 2K17                       |          |          |       |
| Scooby-Doo! Unmasked                                    | 0.01 | 0.09        | 0.0                          | 0.0          | 0.0          | 2005     | 0.0             | Platform              | THQ                        |               | 14.0   | 0.07                           | 0.0      | 0.0      |       |
|                                                         |      |             | Scooby-Doo! Unmasked         |              |              |          |                 | Unknown Developer     |                            | Not Rated     |        | Scooby-Doo! Unmasked           |          |          |       |
| Left Brain Right Brain: Use Both Hands Train Both Sides | 0.01 | 0.16        | 0.0                          | 0.0          | 0.0          | 2007     | 0.0             | Misc                  | 505 Games                  |               | 0.15   | 0.0                            | 0.0      |          |       |
| th Hands Train Both Sides                               |      |             | Parity - Brain Train         |              |              |          |                 | Unknown Developer     |                            | Not Rated     |        | Left Brain Right Brain: Use Bo |          |          |       |
| Zatch Bell! Electric Arena                              | 0.0  | 0.03        | 0.0                          | 0.0          | 0.0          | 2003     | 0.0             | Fighting              | Banpresto                  |               | 0.02   | 0.01                           | 0.0      |          |       |
|                                                         |      |             | Zatch Bell! Electric Arena   |              |              |          |                 | Unknown Developer     |                            | Not Rated     |        | Zatch Bell! Electric Arena     |          |          |       |
| Disney's Chicken Little: Ace In Action                  | 0.0  | 0.06        | 66.0                         | 13.0         | 0.0          | 2006     | 0.0             | Shooter               | Disney Interactive Studios |               | 0.06   | 0.0                            | 0.0      |          |       |
| n Action                                                |      |             | Disney's Chicken Little: Ace |              |              |          |                 | DC Studios            | E                          |               |        | Disney's Chicken Little: Ace I |          |          |       |
| Catwoman                                                |      |             |                              |              |              | 2004     | 3.14            | Action                | Electronic Arts            |               | 1.0    | 0.04                           | 0.01     | 0.0      |       |

creation, and data visualization using Power BI.

```
sqlite> SELECT NAME, PLATFORM, GENRE, RAWG_RATING
...> FROM games_data
...> LIMIT 10;
```

| NAME                                                    | PLATFORM | GENRE     | RAWG_RATING |
|---------------------------------------------------------|----------|-----------|-------------|
| The Incredible Hulk: Ultimate Destruction               | XB       | Action    | 4.12        |
| Datenshi no Amai Yuuwaku x Kaikan Phrase                | DS       | Adventure | 0.0         |
| NBA 2K17                                                | X360     | sports    | 3.33        |
| Scooby-Doo! Unmasked                                    | DS       | Platform  | 4.09        |
| Left Brain Right Brain: Use Both Hands Train Both Sides | DS       | Misc      | 0.0         |
| Zatch Bell! Electric Arena                              | GBA      | Fighting  | 0.0         |
| Disney's Chicken Little: Ace In Action                  | ds       | Shooter   | 3.14        |
| Catwoman                                                | XB       | Action    | 2.03        |
| Rubik's World                                           | wii      | Puzzle    | 0.0         |
| Super Robot Taisen OG: Original Generations Gaiden      | PS2      | Strategy  | 4.83        |

```
sqlite> |
```

Figure 9 Merged data in database

## Step 5: Automating the Workflow

To automate the ETL workflow, a main() function was implemented to execute all pipeline stages in sequence. The process begins by extracting data from the CSV file, followed by retrieving additional game information from the RAWG API using selected game names.

After extraction, the CSV and API datasets are transformed, cleaned, and merged into a unified dataset. The final cleaned data is then loaded into the SQLite database and exported as a CSV file for visualization and further analysis.

This automated workflow simplifies the ETL execution process and allows the entire pipeline to run using a single command.

```
211 # --- Run project ---
212 def main():
213     print("Starting ETL Pipeline...")
214
215     # Extract CSV
216     raw_csv = extract_csv(CSV_FILE_PATH)
217
218     # Use only first 10 unique names for API to avoid slow requests
219     sample_names = (
220         raw_csv["NAME"]
221         .dropna()
222         .astype(str)
223         .str.strip()
224         .unique()[:10]
225         .tolist()
226     )
227
228     # Extract API
229     raw_api = extract_api(sample_names)
230
231     final_df = transform(raw_csv, raw_api)
232
233     load(final_df, DB_PATH)
234
235     final_df.to_csv("final_games.csv", index=False)
236
237     print("Final CSV saved.")
238     print("Done!")
239
240
241 if __name__ == "__main__":
242     main()
243
```

*Figure 10 Workflow Automation*

## 4.2 Visualize using BI dashboard

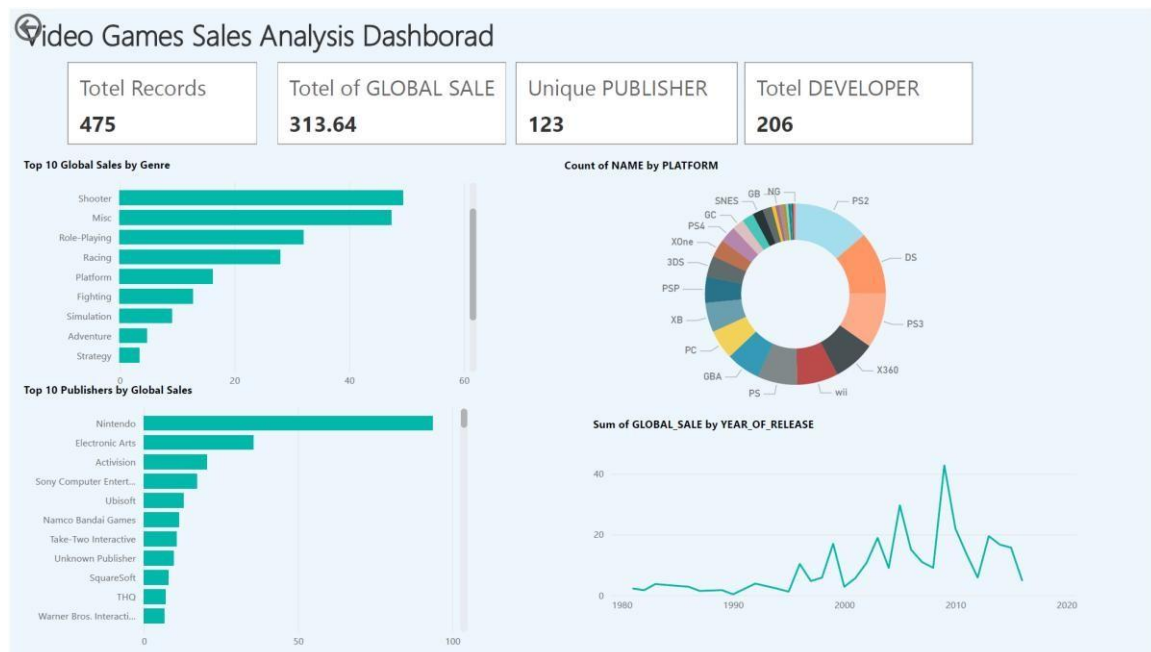


Figure 11 ETL Dashboard

## 4.3 Performance challenges and solutions

During the ETL implementation, some challenges appeared while retrieving data from the RAWG API, such as slow responses and timeout issues. In addition, the CSV and API sources contained missing values, duplicated records, and inconsistent data types.

These issues were addressed by limiting API requests, cleaning missing values, converting columns into suitable data types, and removing duplicate rows using pandas functions. The final dataset was successfully transformed into a clean and structured format ready for storage and dashboard analysis.

## Chapter 5: (Phase 2) Introduction to ELT

## **5.1 Difference between ELT (Extract, Load, Transform) and ETL (Extract ,Transform , load)**

In ETL, the data is collected from different sources, cleaned and transformed first, then loaded into the database or data warehouse.

In ELT, the raw data is collected and loaded directly into the database first. After that, the transformation is done inside the database using SQL.

## **5.2 Why ELT is used**

ELT is used because it is fast, flexible, and suitable for large datasets. It keeps the raw data in the database, so we can check it or transform it again when needed.

In our project, ELT loads the raw video games data first, then cleans and transforms it inside the database for dashboard analysis

## **5.3 Tools used as Datawarehouse and libraries for ELT**

### **1. Programming Language:**

Python: Used to extract and load the raw video games data into the database.

SQL: Used to clean, transform, and query the data inside the data warehouse.

### **2. Python Libraries:**

pandas: Used for reading the CSV file and preparing the raw dataset.

sqlalchemy: Used to connect Python with the PostgreSQL/Neon database.

sqlite3: Used to connect and work with the SQLite database during the extraction and loading process.

### 3.Data Warehouse:

SQLite: Used as a local database to store and manage raw data before loading it into PostgreSQL.

PostgreSQL (Neon): Used as the central data warehouse where raw data is first loaded, then transformed using SQL queries.

### 4.Data Transformation:

SQL Queries inside PostgreSQL (Neon): Used to clean and transform the data directly in the data warehouse.

For example, missing values can be handled using COALESCE, text can be cleaned using TRIM.

## **Chapter 6: ELT Pipeline Design**

## 6.1 Data pipeline architecture (ELT)

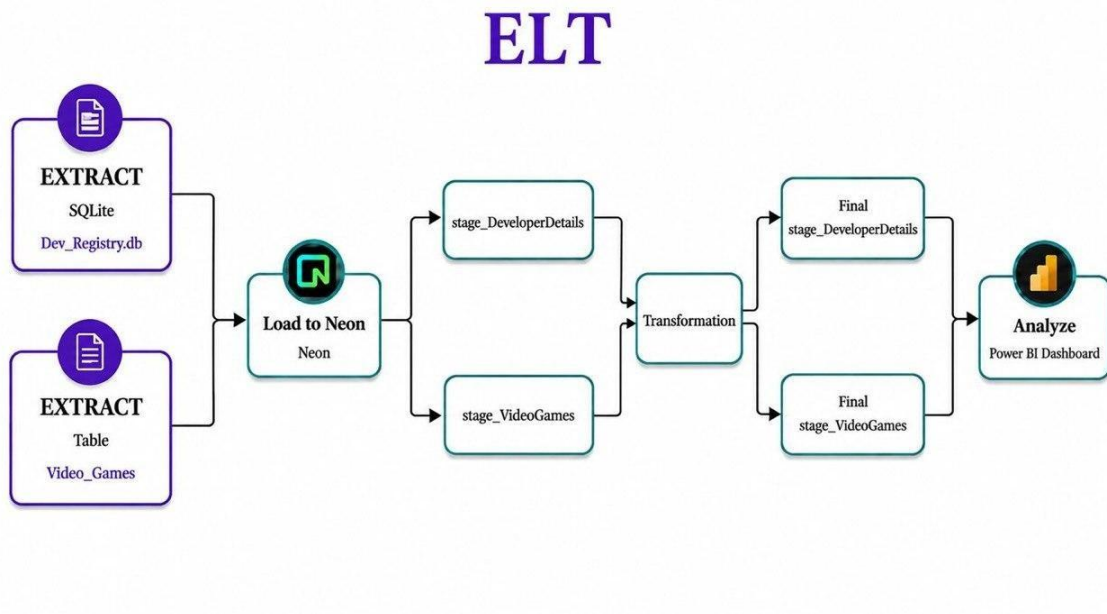


Figure 12 ELT pipeline architecture

## 6.2 Data sources: Collecting raw data

We collected raw gaming data from two distinct sources to capture both developer metadata and performance metrics:

1. SQLite Database: A database containing developer information such as headquarters and foundation years.

```

c:\sqlite>sqlite3
SQLite version 3.51.3 2026-03-13 10:38:09
Enter ".help" for usage hints.
Connected to a transient in-memory database.
Use ".open FILENAME" to reopen on a persistent database.
sqlite> .open Dev_Registry.db
sqlite> .database
main: "" r/w
sqlite> CREATE TABLE IF NOT EXISTS developer_details (
(xl...>     dev_id INTEGER PRIMARY KEY AUTOINCREMENT,
(xl...>     Developer TEXT,
(xl...>     Headquarters TEXT,
(xl...>     Founded_Year INTEGER
(xl...> );

```

Figure 13 Database Creation

```

C:\WINDOWS\system32\CMD - X
sqlite> SELECT * FROM developer_details;
1|Nintendo|Kyoto, Japan|1889
2|Nintendo|Kyoto, Japan|1889
3|Nintendo|1889
4|Ubisoft|Montreuil, France|1986
5|Ubisoft|Montreuil, France|1986
6|Electronic Arts|Redwood City, USA|1982
7|ELECTRONIC ARTS|Redwood City, USA|1982
8|Capcom|Osaka, Japan|1979
9|Capcom|1979
10|Square Enix|Tokyo, Japan|1975
11|Square Enix|Tokyo, Japan|1975
12|Sega|Tokyo, Japan|1968
13|sega|Tokyo, Japan|1968
14|Activision|California, USA|1979
15|Activision|1979
16|Bandai Namco|Tokyo, Japan|2005
17|Bandai Namco|Tokyo, Japan|2005
18|Konami|Tokyo, Japan|1969
19|KONAMI|Tokyo, Japan|1969
20|Take-Two Interactive|New York, USA|1993
21|Take-Two Interactive|1993
22|Sony Interactive Entertainment|San Mateo, USA|1993
23|Sony Interactive Entertainment|San Mateo, USA|1993
24|Bethesda Softworks|Maryland, USA|1986
25|bethesda softworks|Maryland, USA|1986
26|Microsoft Game Studios|Washington, USA|2000
27|Microsoft Game Studios||2000
28|Rockstar Games|New York, USA|1998
29|Rockstar Games|New York, USA|1998
30|Valve Corporation|Washington, USA|1996
31|VALVE CORPORATION|Washington, USA|1996
32|BioWare|Edmonton, Canada|1995
33|BioWare||1995
34|CD Projekt Red|Warsaw, Poland|2002
35|CD Projekt Red|Warsaw, Poland|2002
36|FromSoftware|Tokyo, Japan|1986
37|FromSoftware|Tokyo, Japan|1986
38|Bungie|Washington, USA|1991
39|Insomniac Games|California, USA|1994
40|Insomniac Games|California, USA|1994
41|Naughty Dog|California, USA|1984
42|NAUGHTY DOG|California, USA|1984

```

Figure 14 First 42 row of the data

To extract it, we use the sqlite3 library in Python to connect to the database and pandas to read the SQL query results:

```

# 1. Extract from SQLite Database
conn_sqlite = sqlite3.connect(r"C:\sqlite\Dev_Registry.db")
df_devs = pd.read_sql_query("SELECT * FROM developer_details", conn_sqlite)
conn_sqlite.close()

```

Figure 15 Extract data from SQLite

2. CSV file (Video\_Games.csv): This file contains a dataset of 500

```

# 2. Extract from CSV
csv_path = r"C:\Users\jsdsj\venv\Video_Games.csv"
df_csv = pd.read_csv(csv_path)

```

Figure 16 Extract data from csv file.

video game records, including attributes like Platform, Genre, and Global Sales.

By combining these two sources, we provide a comprehensive view of both developer profiles and game performance for our analysis.

### 6.3 Load the raw data in datawarehouse

After extracting the data, we established a connection to our cloud data warehouse on Neon PostgreSQL. We used the SQLAlchemy library in Python to create an engine using our Neon connection string.

- `stage_DeveloperDetails`: for the data extracted from the SQLite database.
- `stage_VideoGames`: for the dataset loaded from the CSV file.

By using the `to_sql` function with our Neon engine, we ensured that the raw data was successfully migrated from our local environment to the cloud

```
# 3. Connect to Neon PostgreSQL
NEON_URL = "postgresql://neondb_owner:npg_g3z8cRGSVlmh@ep-odd-darkness-ap13lky9-pooler.c-7.us-east-1.aws.neon.tech/resp
engine = create_engine(NEON_URL)

# 4. Load both into Neon (Staging Tables)
df_devs.to_sql("stage_DeveloperDetails", engine, if_exists="replace", index=False)
df_csv.to_sql("stage_VideoGames", engine, if_exists="replace", index=False)

print("Data loaded into the staging tables successfully!")
```

Figure 17 establishing connection to Neon cloud data warehouse using SQLAlchemy.

```
(.venv) PS C:\Users\jsdsj\venv> python project.py
Data loaded into the staging tables successfully!
```

Figure 18 output confirming successful script execution.

for centralized storage.

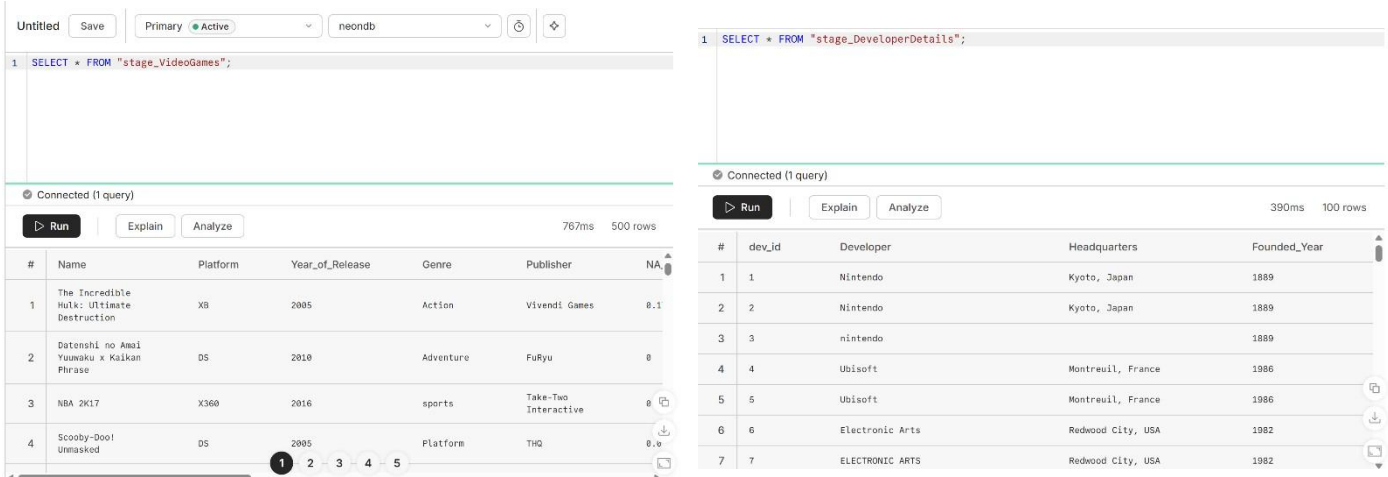


Figure 19 Staging tables

## 6.4 Transform: SQL based transformation

After extracting and loading the raw data into the Neon PostgreSQL staging tables, , we applied SQL cleaning and transformation directly within the Neon PostgreSQL environment. Using the SQL Editor, we moved the data from staging tables to final tables while ensuring data quality through specific SQL functions.

Table 1: final\_VideoGames:

This final table was created in Neon to store the cleaned version of the video games dataset.

```

1 DROP TABLE IF EXISTS final_VideoGames;
2 CREATE TABLE final_VideoGames (
3     game_id SERIAL PRIMARY KEY,
4     Name TEXT,
5     Platform TEXT,
6     Year_of_Release INTEGER,
7     Genre TEXT,
8     Publisher TEXT,
9     Global_Sales REAL
10 );

```

*Figure 20 Final Table for VideoGames Transformations*

performed:

SELECT DISTINCT: Used to remove any duplicate game records.

TRIM(): Used to remove extra spaces from the game Names and Platforms.

UPPER(): Used on the Platform column to standardize it (e.g., PS4, PC).

CAST(... AS INTEGER): Used to convert the Year\_of\_Release into a numeric format for sorting.

COALESCE(): This function was used for multiple purposes:

- To replace missing Year\_of\_Release values with 0.
- To handle missing Genre entries by replacing them with the default category 'Miscellaneous'.
- To replace empty Publisher fields with 'Unknown' and null Global\_Sales with 0.0.

WHERE "Name" IS NOT NULL: Used to ensure no incomplete records without names are inserted.

```
INSERT INTO final_videogames (name, platform, year_of_release, genre, publisher, global_sales)
SELECT DISTINCT
  TRIM("Name") AS name,
  UPPER(TRIM("Platform")) AS platform,
  COALESCE(CAST("Year_of_Release" AS INTEGER), 0) AS year_of_release,
  COALESCE("Genre", 'Miscellaneous') AS genre,
  COALESCE("Publisher", 'Unknown') AS publisher,
  COALESCE("Global_Sales", 0.0) AS global_sales
FROM "stage_VideoGames"
WHERE "Name" IS NOT NULL
```

*Figure 21 Transformations for videoGames*

| 50 rows • 308ms          |                           |               |                           |              |                           |                     |
|--------------------------|---------------------------|---------------|---------------------------|--------------|---------------------------|---------------------|
| <input type="checkbox"/> | Name text                 | Platform text | Year_of_Release double... | Genre text   | Publisher text            | NA_Sales double pre |
| <input type="checkbox"/> | The Incredible Hulk: U... | XB            | 2005                      | Action       | Vivendi Games             | 0.17                |
| <input type="checkbox"/> | Datenshi no Amai Yuuwa... | DS            | 2010                      | Adventure    | FuRyu                     | 0                   |
| <input type="checkbox"/> | NBA 2K17                  | X360          | 2016                      | sports       | Take-Two Interactive      | 0.09                |
| <input type="checkbox"/> | Scooby-Doo! Unmasked      | DS            | 2005                      | Platform     | THQ                       | 0.07                |
| <input type="checkbox"/> | Left Brain Right Brain... | DS            | 2007                      | Misc         | 505 Games                 | 0.15                |
| <input type="checkbox"/> | Zatch Bell! Electric A... | GBA           | 2003                      | Fighting     | Banpresto                 | 0.02                |
| <input type="checkbox"/> | Disney's Chicken Littl... | ds            | 2006                      | Shooter      | Disney Interactive Stu... | 0.06                |
| <input type="checkbox"/> | Catwoman                  | XB            | 2004                      | Action       | Electronic Arts           | 0.04                |
| <input type="checkbox"/> | Rubik's World             | wii           | 2008                      | Puzzle       | Game Factory              | 0.12                |
| <input type="checkbox"/> | Super Robot Taisen OG...  | PS2           | 2007                      | Strategy     | Banpresto                 | 0                   |
| <input type="checkbox"/> | J-League Pro Soccer CL... | PSP           | 2011                      | sports       | Sega                      | 0                   |
| <input type="checkbox"/> | Maji de Manabu: LEC de... | DS            | 2008                      | Misc         | Square Enix               | 0                   |
| <input type="checkbox"/> | The Sims 3: Town Life ... | PC            | 2011                      | Simulation   | Electronic Arts           | 0.11                |
| <input type="checkbox"/> | The Elder Scrolls IV: ... | PC            | 2006                      | Role-Playing | Take-Two Interactive      | 0.01                |
| <input type="checkbox"/> | Let's Cheer               | X360          | 2011                      | Misc         | Take-Two Interactive      | 0.12                |
| <input type="checkbox"/> | Rayman 2: Revolution      | PS2           | 2000                      | Platform     | Ubisoft                   | 0.15                |
| <input type="checkbox"/> | Captain America: Super... | PS3           | 2011                      | Action       | Sega                      | 0.1                 |
| <input type="checkbox"/> | NBA Live 08               | PS2           | 2007                      | Sports       | Electronic Arts           | 0.59                |
| <input type="checkbox"/> | LEGO Star Wars II: The... | XB            | 2006                      | Action       | LucasArts                 | 0.33                |
| <input type="checkbox"/> | Minecraft: Story Mode     | X360          | 2015                      | Adventure    | Mojang                    | 0.48                |

Figure 22 before appley transformations

| <input type="checkbox"/> | game_id serial | name text                 | platform text | year_of_release integer | genre text   | publisher text   |
|--------------------------|----------------|---------------------------|---------------|-------------------------|--------------|------------------|
| <input type="checkbox"/> | 1              | TV Anime Idolmaster: C... | PS3           | 2016                    | Misc         | Unknown          |
| <input type="checkbox"/> | 2              | Rhythm Heaven: The Bes... | 3DS           | 2015                    | Misc         | Nintendo         |
| <input type="checkbox"/> | 3              | All Grown Up!: Game Bo... | GBA           | 2004                    | misc         | Unknown          |
| <input type="checkbox"/> | 4              | From Russia With Love     | XB            | 2005                    | Action       | Electronic Arts  |
| <input type="checkbox"/> | 5              | Super Monkey Ball Delu... | XB            | 2005                    | Misc         | Sega             |
| <input type="checkbox"/> | 6              | Fantastic 4               | XB            | 2005                    | Action       | Activision       |
| <input type="checkbox"/> | 7              | Hulk                      | XB            | 2003                    | Action       | Universal Intera |
| <input type="checkbox"/> | 8              | Desktop Tower Defense     | DS            | 2009                    | Strategy     | THQ              |
| <input type="checkbox"/> | 9              | WSC Real 11: World Sno... | PS3           | 2011                    | Sports       | Koch Media       |
| <input type="checkbox"/> | 10             | Goat Simulator            | XONE          | 2016                    | Simulation   | Koch Media       |
| <input type="checkbox"/> | 11             | Pocket Pets               | DS            | 2007                    | Simulation   | O3 Entertainment |
| <input type="checkbox"/> | 12             | Just Dance 4              | X360          | 2012                    | Misc         | Ubisoft          |
| <input type="checkbox"/> | 13             | Cabela's North America... | PSP           | 2010                    | Sports       | Activision       |
| <input type="checkbox"/> | 14             | Teenage Mutant Ninja T... | DS            | 2009                    | Action       | Ubisoft          |
| <input type="checkbox"/> | 15             | Black Bass with Blue M... | PS            | 1999                    | Sports       | Starfish         |
| <input type="checkbox"/> | 16             | Pirates of the Caribbe... | PSP           | 2006                    | Adventure    | Disney Interacti |
| <input type="checkbox"/> | 17             | Bass Hunter 64            | N64           | 1999                    | Sports       | Take-Two Interac |
| <input type="checkbox"/> | 18             | Monster Hunter Frontie... | X360          | 2010                    | Role-Playing | Capcom           |
| <input type="checkbox"/> | 19             | NBA 2K11                  | X360          | 2010                    | Action       | Take-Two Interac |
| <input type="checkbox"/> | 20             | Kirby's Adventure         | NES           | 1993                    | Platform     | Nintendo         |

Figure 23 after appley transformations

- Table 2: final\_DeveloperDetails:

A second transformation was applied to the developers' data to ensure consistency across the database.

```

1 DROP TABLE IF EXISTS final_DeveloperDetails;
2
3 CREATE TABLE final_DeveloperDetails (
4     id SERIAL PRIMARY KEY,
5     Developer TEXT,
6     Headquarters TEXT,
7     Founded_Year INTEGER
8 );

```

*Figure 24 Final table for DeveloperDetails*

Transformations performed:

INITCAP(): Used to capitalize the first letter of each Developer name.

TRIM(): Used to clean the developer names from leading or trailing spaces.

COALESCE(): Used to replace missing Headquarters data with the value 'Unknown'.

SELECT DISTINCT: Used to ensure each developer is only listed once in the final table.

WHERE "Developer" IS NOT NULL: Used to ensure no incomplete records without Developer are inserted.

```

1 INSERT INTO final_developerdetails (developer, headquarters, founded_year)
2 SELECT DISTINCT
3     INITCAP(TRIM("Developer")) AS developer,
4     COALESCE("Headquarters", 'Unknown') AS headquarters,
5     "Founded_Year"
6 FROM "stage_DeveloperDetails"
7 WHERE "Developer" IS NOT NULL

```

*Figure 25 Transformations for DeveloperDetails*

| dev_id bigint | Developer text       | Headquarters text | Founded_Year bigint |
|---------------|----------------------|-------------------|---------------------|
| 1             | Nintendo             | Kyoto, Japan      | 1889                |
| 2             | Nintendo             | Kyoto, Japan      | 1889                |
| 3             | nintendo             | NULL              | 1889                |
| 4             | Ubisoft              | Montreuil, France | 1986                |
| 5             | Ubisoft              | Montreuil, France | 1986                |
| 6             | Electronic Arts      | Redwood City, USA | 1982                |
| 7             | ELECTRONIC ARTS      | Redwood City, USA | 1982                |
| 8             | Capcom               | Osaka, Japan      | 1979                |
| 9             | Capcom               | NULL              | 1979                |
| 10            | Square Enix          | Tokyo, Japan      | 1975                |
| 11            | Square Enix          | Tokyo, Japan      | 1975                |
| 12            | Sega                 | Tokyo, Japan      | 1960                |
| 13            | sega                 | Tokyo, Japan      | 1960                |
| 14            | Activision           | California, USA   | 1979                |
| 15            | Activision           | NULL              | 1979                |
| 16            | Bandai Namco         | Tokyo, Japan      | 2005                |
| 17            | Bandai Namco         | Tokyo, Japan      | 2005                |
| 18            | Konami               | Tokyo, Japan      | 1969                |
| 19            | KONAMI               | Tokyo, Japan      | 1969                |
| 20            | Take-Two Interactive | New York, USA     | 1993                |
| 21            | Take-Two Interactive | NULL              | 1993                |

Figure 26 before appley transformations

| id serial | developer text            | headquarters text      | founded_year integer |
|-----------|---------------------------|------------------------|----------------------|
| 1         | Bungie                    | Unknown                | 1991                 |
| 2         | Capcom                    | Unknown                | 1979                 |
| 3         | Bethesda Softworks        | Maryland, USA          | 1986                 |
| 4         | Media Molecule            | Guildford, UK          | 2006                 |
| 5         | Electronic Arts           | Redwood City, USA      | 1982                 |
| 6         | Atlas                     | Tokyo, Japan           | 1986                 |
| 7         | Platiniumgames            | Osaka, Japan           | 2006                 |
| 8         | Larian Studios            | Unknown                | 1996                 |
| 9         | Bandai Namco              | Tokyo, Japan           | 2005                 |
| 10        | Codemasters               | Southam, UK            | 1986                 |
| 11        | Id Software               | Unknown                | 1991                 |
| 12        | Riot Games                | California, USA        | 2006                 |
| 13        | Netease                   | Hangzhou, China        | 2001                 |
| 14        | Guerrilla Games           | Amsterdam, Netherlands | 2000                 |
| 15        | Activision                | California, USA        | 1979                 |
| 16        | Santa Monica Studio       | Unknown                | 1999                 |
| 17        | Rare                      | Unknown                | 1985                 |
| 18        | Take-Two Interactive      | New York, USA          | 1993                 |
| 19        | Insomniac Games           | California, USA        | 1994                 |
| 20        | Sony Interactive Enter... | San Mateo, USA         | 1993                 |
| 21        | Id Software               | Texas, USA             | 1991                 |

Figure 27 after appley transformations

## Chapter 7: Implementation & Results

### 7.1 Execution of ELT workflow

The ELT pipeline was executed successfully based on the architecture and workflow designed in Chapter 6. The implementation followed three main stages: extraction, loading, and transformation. The workflow was developed using Python, PostgreSQL (Neon), SQLAlchemy, pandas, sqlite3, and SQL queries executed in PostgreSQL (Neon).

#### Tools and Technologies Used:

Python (Visual Studio Code): Used to implement and automate the ELT workflow.

Pandas: Used for extracting and reading data from CSV files and SQLite databases. sqlite3: Used to connect with the local SQLite database during extraction.

SQLAlchemy: Used to establish a connection between Python and Neon PostgreSQL.

PostgreSQL (Neon) : Used as the cloud data warehouse for staging, transformation, and final storage using SQL queries.

### **7.1.1 Extract**

In the extraction phase, raw gaming data was collected from two different sources:

A local SQLite database containing developer information such as developer names, headquarters, and foundation years.

A CSV file named Video\_Games.csv containing video game records including platform, genre, publisher, and global sales.

The extraction process was implemented using: `sqlite3.connect()` to connect to the SQLite database. `pd.read_sql_query()` to extract database records into pandas DataFrames.

`pd.read_csv()` to load the CSV dataset into Python.

### **7.1.3 Load**

After extraction, the raw datasets were loaded directly into Neon PostgreSQL staging tables using SQLAlchemy and the pandas `to_sql()` function.

The following staging tables were created:

`stage_DeveloperDetails` `stage_VideoGames`

The loading phase migrated the raw data from the local environment into the cloud data warehouse successfully. Console output messages confirmed that the loading process completed without errors.

#### **7.1.4 Transform (SQL)**

After loading the raw data into PostgreSQL (Neon) staging tables, SQL transformations were executed directly inside Neon PostgreSQL using internal dieter.

Several SQL functions were applied to clean and standardize the data, including:

SELECT DISTINCT to remove duplicate records.

TRIM() to clean extra spaces from text columns.

UPPER() to standardize platform names.

COALESCE() to replace missing values with default values.

CAST(... AS INTEGER) to convert release years into numeric format.

The cleaned data was then inserted into the final tables:

final\_VideoGames final\_DeveloperDetails

#### **7.1.5 Execution Result**

The ELT workflow completed successfully without runtime errors. The final PostgreSQL tables contained cleaned, structured, and standardized gaming data ready for dashboard analysis and visualization.

The execution results confirmed that:

Data extraction completed successfully.

Raw data was loaded correctly into PostgreSQL staging tables.

SQL transformations improved data quality and consistency.

Final tables were generated successfully for reporting and BI dashboard visualization.

## 7.2 Visualize using BI dashboard

### Video Games Sales Analysis Dashborad



Figure 28 Final BI Dashboard visualizing the processed data

## 7.3 Automation of ELT pipeline using Task scheduler

I used ChatGPT promote to help understand Task scheduler and create one

Figure 29 AI integration

Explain Task Scheduler in Windows, and how to create it using task schedule? explain all steps for scheduling Python script automatically every day at 11:59 PM for ELT/data pipeline project.

### What is Windows Task Scheduler?

Windows Task Scheduler is a built-in tool in Windows that allows users to run programs or scripts automatically at a specific time or event. It is very useful in ELT or data pipeline projects because it can run Python scripts without opening them manually every day.

For example in ELT projects Task Scheduler can:

- Run data extraction scripts every night
- Update databases automatically
- Generate reports daily
- Automate data cleaning and transformation

In our case we will use it to run a Python script automatically every day at 11:59 PM.

---

## Steps to Schedule a Python Script in Task Scheduler

### Step 1: Prepare the Python Script

Make sure your Python script works correctly before scheduling it.

Example:

```
</> Python  
print("ELT Pipeline Running")
```



Save the file for example as:

```
pipeline.py
```



Example location:

```
C:\ELT_Project\pipeline.py
```



---

## Step 2: Find Python.exe Path

Task Scheduler needs the location of Python.

How to find it:

1. Open Command Prompt
2. Type:

```
<> Bash  
where python
```



Example output:

```
C:\Users\Sara\AppData\Local\Programs\Python\Python312\python.exe
```



Copy this path because we will use it later.

---

## Step 3: Open Task Scheduler

1. Press Windows Search
2. Type:

```
Task Scheduler
```



3. Open the application
- 

## Step 4: Create a Basic Task

On the right side click:

```
Create Basic Task
```



---

## Step 5: Name the Task

Write:

- Name:

```
ELT Pipeline Automation
```



- Description:

```
Run Python ELT pipeline every day automatically
```



Then click:

```
Next
```



## Step 6: Choose Trigger

Trigger means when the task will run.

1. Select:

Daily 

2. Click:



Next 

## Step 7: Set Time

Choose:

11:59:00 PM 

Set:

- Recur every:

1 day 

Then click:

Next 

## Step 8: Choose Action

Select:

Start a Program 

Then click:



Next 

## Step 9: Add Python Program

In Program/script:

Paste Python path.

Example:

C:\Users\Sara\AppData\Local\Programs\Python\Python312\python.exe 

### In Add arguments:

Write the script path inside quotes.

Example:

"C:\ELT\_Project\pipeline.py"



### In Start in:

Write the folder location only.

Example:

C:\ELT\_Project



Then click:

Next



## Step 10: Finish the Task

Check the information then click:

Finish



Now the Python ELT script will run automatically every day at 11:59 PM.

## How to Test the Task

Before waiting until 11:59 PM you should test it.

1. Open Task Scheduler
2. Find your task
3. Right click the task
4. Press:

Run



If everything is correct the Python script will execute automatically.

My implementation **Task Scheduler**.

- Scheduling: I set a daily trigger to run the Python script (Project.py) at a specific time.
- Configuration: I used the Python interpreter from the virtual environment (venv) to ensure all libraries work correctly.

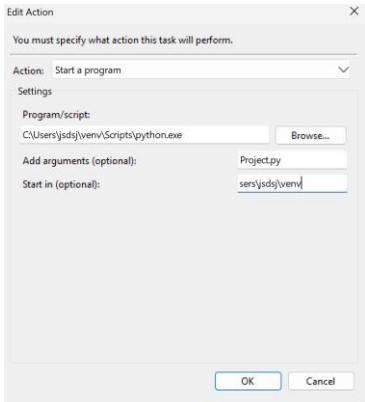


Figure 30 Configuring Task Scheduler action with the virtual environment.

- Verification: The task was successful, showing the status (0x0), which confirms that the data was transferred to Neon PostgreSQL effectively.

| Last Run Time        | Last Run Result               |
|----------------------|-------------------------------|
| 5/5/2026 10:29:22 PM | The operation completed succe |
| 5/6/2026 4:36:14 PM  | The operation completed succe |

Figure 31 Task Scheduler history confirming successful run .



Figure 32 database updated successfully following the automated run

## 7.4 Comparison of ETL vs. ELT (speed, scalability, efficiency)

| Performance Compariso | ETL Pipeline  | ELT Pipeline  |
|-----------------------|---------------|---------------|
| Speed                 | 14.529 sec    | 15.03 sec     |
| Scalability           | Less scalable | More scalable |
| Efficiency            | Lower         | Higher        |

*Table 1 ETL vs ELT*

### **Speed**

**ETL:** The ETL pipeline completed in 14.529 seconds, as local pandas transformations and API merging were performed before loading into SQLite.

**ELT:** The ELT pipeline took 15.03 seconds for extraction and loading only. The slight increase is due to network latency when connecting to the Neon PostgreSQL cloud warehouse.

### **Scalability**

**ETL:** Less scalable since it depends on local RAM to process the full dataset, which may not handle large or complex datasets efficiently as data volume grows.

**ELT:** More scalable as it leverages the power of cloud-based PostgreSQL (Neon) to handle large datasets natively without overloading the local machine.

## **Efficiency**

**ETL:** Lower efficiency due to multiple explicit transformation steps written in Python, which increases code complexity and maintenance overhead.

**ELT:** Highly efficient because Python handles only extraction and loading, while all transformations are executed inside Neon using SQL, reducing code complexity and enabling easier auditing and modification.

## **7.5 Conclusion**

This project successfully demonstrated the design and implementation of both **ETL** and **ELT** pipelines applied to a video games dataset. By integrating data from a CSV file, the **RAWG API**, and a local **SQLite** database, the project

provided a practical comparison of how architectural choices impact performance and scalability.

In the **ETL phase**, data was transformed using **Python and pandas** before being loaded into SQLite. While this approach allowed for granular control over data quality, it faced limitations in speed and scalability due to local memory and CPU constraints. Conversely, the **ELT phase** leveraged the power of the **Neon PostgreSQL** cloud warehouse. By loading raw data directly into staging tables and performing transformations using **SQL**, the pipeline achieved higher efficiency, easier maintainability, and better auditability.

The automation of these pipelines using **Windows Task Scheduler** further ensured a reliable, intervention-free data flow. Ultimately, this project reinforces that while ETL is effective for controlled, smaller workloads, ELT is the superior choice for modern, scalable cloud environments. This hands-on experience highlights the critical importance of selecting the right architecture based on specific use cases and data requirements in real-world data engineering.

## 7.6 References (IEEE format)

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